

Waïss Azizian | ML Researcher

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- Ph.D. in Optimization & Machine Learning
- 12 peer-reviewed publications including NeurIPS, ICML, ICLR, COLT, AISTATS, and JMLR; 600+ citations.

Education

Université Grenoble Alpes

Ph.D., Supervisors: F. Iutzeler, J. Malick, P. Mertikopoulos

Grenoble, France

March 2022 - June 2026

- Studied the long-run behavior of stochastic optimization algorithms on non-convex landscapes: characterization through large deviations theory.
- Developed and analyzed distributionally robust optimization methods

École Normale Supérieure Paris-Saclay

M.Sc. in Machine Learning (MVA)

Graduated with highest honors

Saclay, France

2020 – 2021

École Normale Supérieure

Major in Mathematics & Computer Science

Paris, France

2017 – 2020

Publications

Conferences: 9 conference papers at NeurIPS/ICML/ICLR/COLT/AISTATS

Journals: 3 journal publications, incl. JMLR

Citations: 600+ (Google Scholar)

Selected publications.....

- “What is the long-run distribution of SGD? A large deviations analysis”, ICML 2024
- “How Does the Pretraining Distribution Shape In-Context Learning?”, ICML 2026

Research and industry experience

Morgan Stanley ML Research

PhD ML Research Intern, Supervisor: Ali Hasan

New York, USA

Aug 2025 – Oct 2025

- Investigated in-context learning mechanisms of LLMs and their scaling behavior under distributional shifts
- Improved internal option-pricing reinforcement learning pipeline
- Developed reproducible research infrastructure (GPU cluster, multi-GPU with JAX)

Apple Machine Learning Research

PhD ML Research Intern, Supervisor: Marco Cuturi

Paris, France

Feb 2025 – May 2025

- Evaluated the robustness of uncertainty quantification methods for Large Language Models
- Built probing pipelines on model activations, including sparse and multi-task classifiers.
- Found that learned directions are task-specific with minimal overlap, highlighting limits of activation-based signals.

Mila – Québec AI Institute

Research Intern, Supervisor: Simon Lacoste-Julien

Montreal, Canada

Mar – Jul 2019

- Studied and improved game-theoretic optimization algorithms for ML (min-max / GAN-style settings)

Technical skills

- *Deep Learning:* PyTorch, JAX/Flax, vLLM; experiment pipelines.
- *Research Tools:* Git, Hydra, Docker, Singularity.
- *High-Performance Computing:* cluster infrastructure (SLURM, OAR), distributed computing (Ray).
- *Mathematical expertise:* Probability, statistics, optimization

Research communication

- Presented research in numerous invited seminars for various academic and industry audiences.
- Awarded Prix Dodu for best student talk at SMAI MODE 2026.
- Undergraduate & graduate teaching, including statistics, numerical optimization and Python/Numpy labs.